Lecture 10. Export Subsidies in Agriculture and High-Technology Industries

Lectures on International Trade

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Outline

Subsidies

- WTO Goals on Agricultural Export Subsidies
- Agricultural Export Subsidies in a Small Home Country
- Agricultural Export Subsidies in a Large Home Country
- Agricultural Production Subsidies
- High-Technology Export Subsidies

Introduction

- On July 21, 2008, representatives of the 152 countries belonging to the WTO met in Geneva, Switzerland to discuss reforms of the world trading system. Like earlier meetings in Seattle (1999), and Cancún, Mexico (2003), and Hong Kong (2005), this meeting was marked by large-scale protests.
- The first goal of this chapter is to explain agricultural subsidy policies.
- The second goal of the chapter is to examine how export subsidies can be used strategically by governments to bolster domestic companies and industries.

WTO Goals on Agricultural Export Subsidies

Agricultural Export Subsidies

- An **export subsidy** is payment to firms for every unit exported (either a fixed amount or a fraction of the sales price). Governments give subsidies to encourage domestic firms to produce more in particular industries.
- Europe maintains a system of agricultural subsidies known as the **Common Agricultural Policy (CAP)**.
- Other countries maintain similarly generous subsidies. For example, the U.S. pays cotton farmers to grow more cotton and subsidizes agribusiness and manufacturers to buy the American cotton.

WTO Goals on Agricultural Export Subsidies, cont'

Hong Kong WTO Meeting, December 2005

- **Indirect Subsidies**. Included in the Hong Kong export subsidy agreement is the parallel elimination of indirect subsidies to agriculture.
- **Domestic Farm Supports**. Another item mentioned in the Hong Kong agreement is domestic farm supports, which refers to any assistance given to farmers, even if it is not directly tied to exports.
- **Cotton Subsidies**. Finally, export subsidies in cotton received special attention because that crop is exported by many low-income African countries and is highly subsidized in the United States.

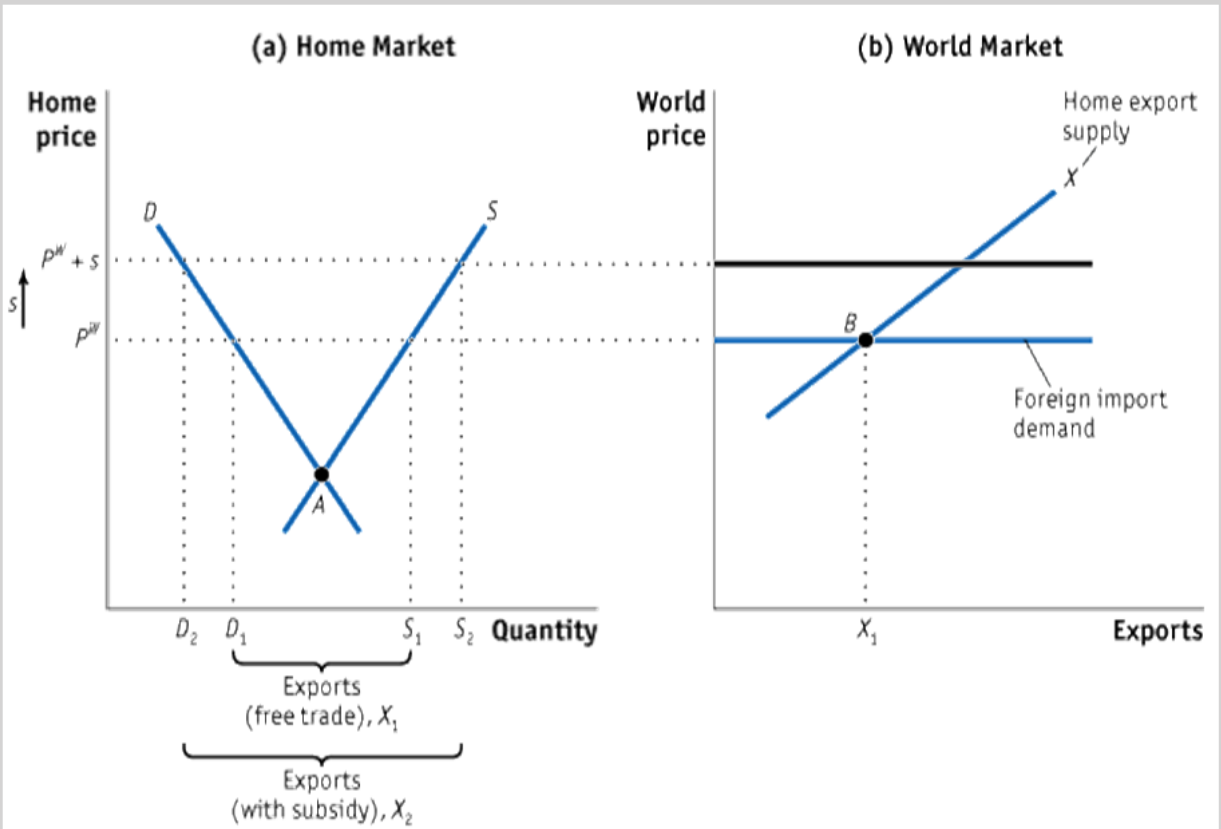
WTO Goals on Agricultural Export Subsidies, cont'

- Issues related to export subsidies were also discussed at the 2005 Hong Kong meeting, in addition to the elimination of the subsidies themselves. One of these issues is the use of tariffs as a response to other countries' use of subsidies.
- **Tariffs in Agriculture** Export subsidies applied by large countries depress world prices, so that exporting countries can expect tariffs to be imposed on the subsidized products when they are imported by other countries.
- There was also an agreement to discuss opening trade in **service sectors**, which would benefit the industrial countries and their large service industries.
- Finally, there was agreement to allow 97% of imported products from the world's 50 least developed countries (LDCs) to enter WTO member markets tariff-free and duty-free.

Agricultural Export Subsidies in a Small Home Country

Impact of an Export Subsidy in a Small Economy

FIGURE 10-1 (1 of 2) **Export Subsidy for a Small Country**



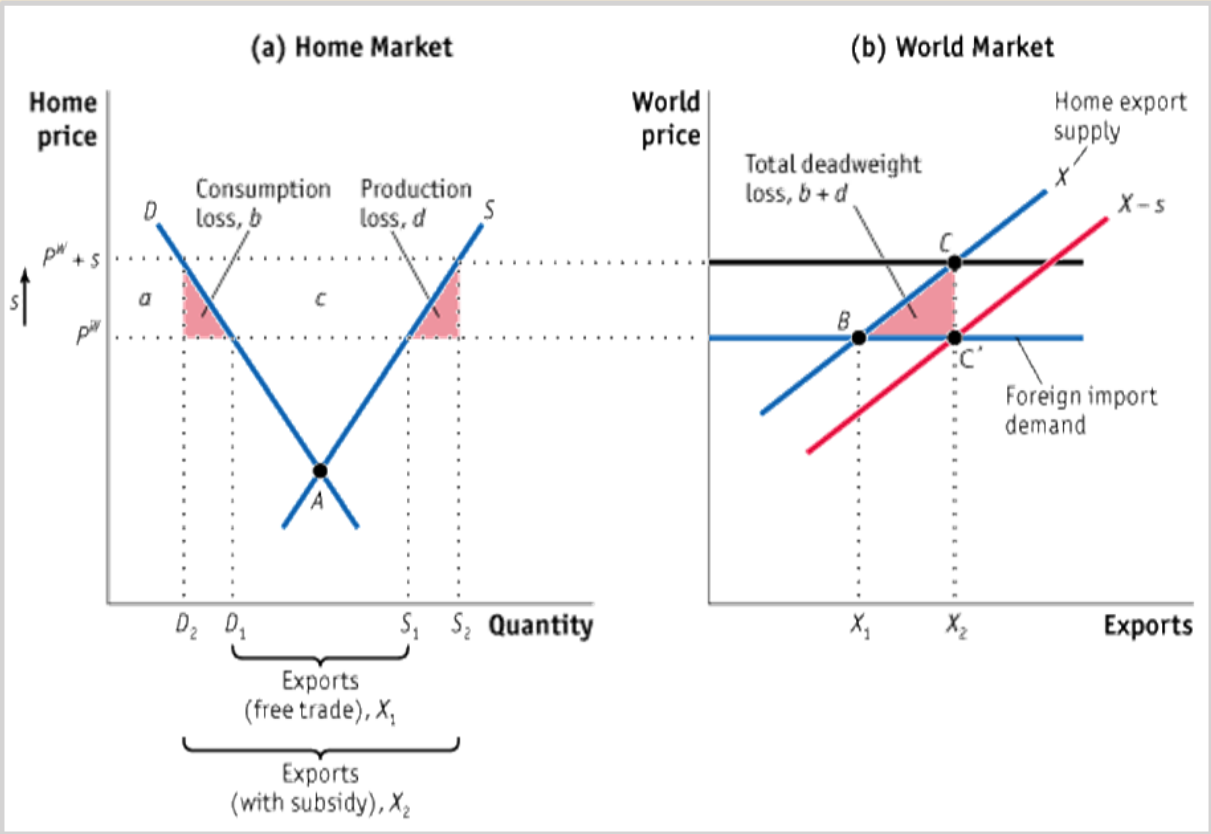
Applying a subsidy of *s* dollars per unit exported will increase the price that Home exporters receive from P^W to $P^W + s$.

As a result, the domestic price of the similar good will also rise by that amount. This price rise leads to an increase in Home quantity supplied from S_1 to S_2 and a decrease in Home quantity demanded from D_1 to D_2, in panel (a).

Agricultural Export Subsidies in a Small Home Country, cont'

Impact of an Export Subsidy in a Small Economy

FIGURE 10-1 (2 of 2) **Export Subsidy for a Small Country (continued)**



Exports rise as a result of the subsidy, from X_1 to X_2 in panel (b).

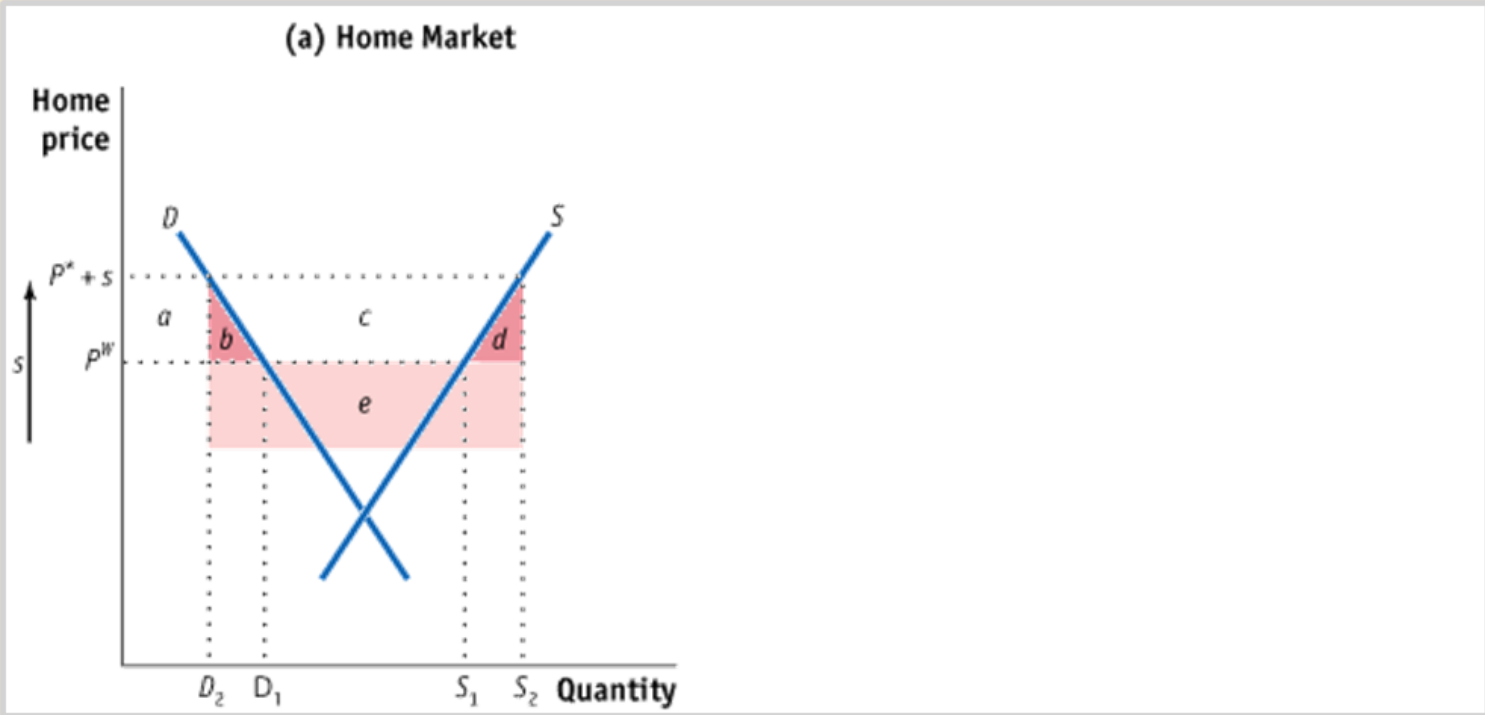
The Home export supply curve shifts down by exactly the amount of the subsidy since the marginal cost of a unit of exports decreases by exactly *s*.

As in the case of a tariff, the deadweight loss as a result of the subsidy is the triangle (*b* + *d*), the sum of consumer loss *b* and producer loss *d*.

Agricultural Export Subsidies in a Large Home Country

Impact of an Export Subsidy in a Large Economy

FIGURE 10-2 (1 of 2) **Export Subsidy for a Large Country**

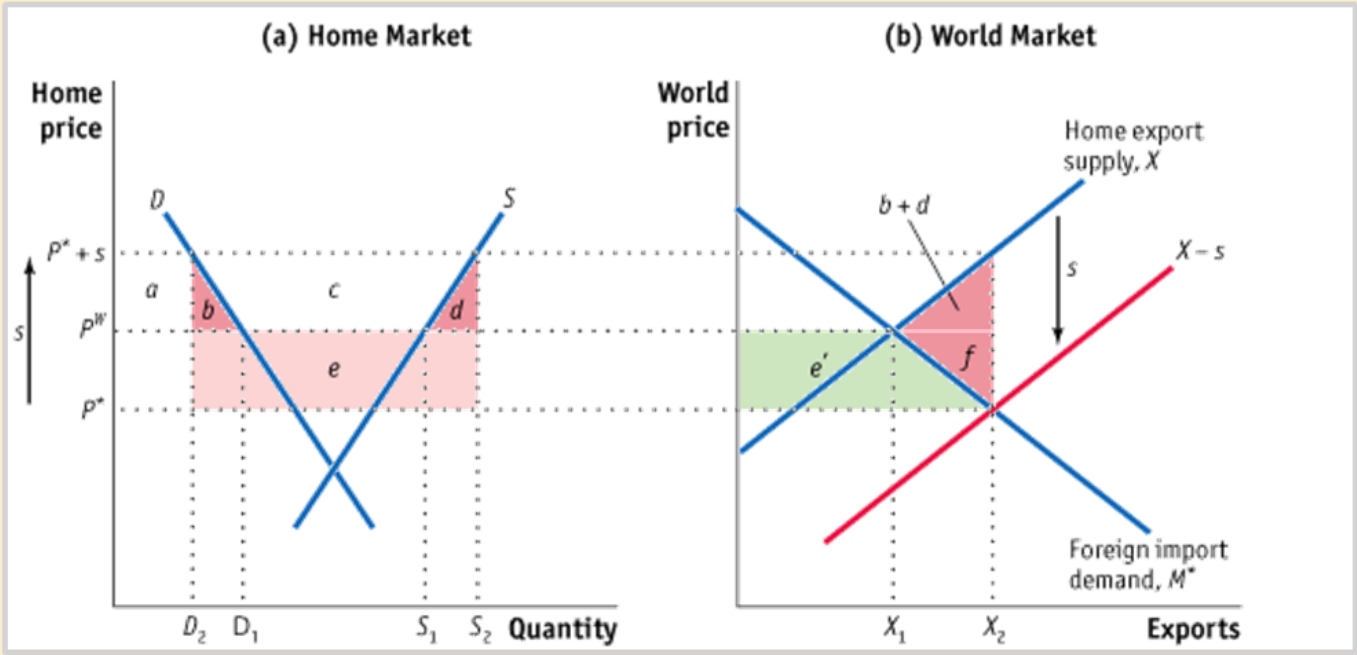


Panel (a) shows the effects of the subsidy at Home. The Home price increases from P^W to $P^* + s$, Home quantity demanded decreases from D_1 to D_2, and Home quantity supplied increases from S_1 to S_2. The deadweight loss for Home is the area of triangle ($b + d$), but Home also has a terms-of-trade loss of area e.

Agricultural Export Subsidies in a Large Home Country

Impact of an Export Subsidy in a Large Economy

FIGURE 10-2 (2 of 2) **Export Subsidy for a Large Country (continued)**



In the world market, the Home subsidy shifts out the export supply curve from X to $X - s$, reflecting the lower marginal cost of exports. As a result, the world price falls from P^W to P^*. The Foreign country gains the consumer surplus area e', so the world deadweight loss due to the subsidy is the area $(b + d + f)$. The extra deadweight loss f arises because only a portion of the Home terms-of-trade loss is a Foreign gain.

WTO Agreement on Subsidies

Who Gains and Who Loses?

- We return to the agreements of the Hong Kong meeting of the WTO in December 2005 and ask: Which countries will gain and which will lose when export subsidies (including the"indirect" subsidies like food aid) are eliminated by 2013?
- **Gains** Current agricultural exporters will gain from the rise in world prices as agricultural subsidies by the industrialized countries-especially Europe and the United States-are eliminated.
- **Losses** The food-importing countries, typically the poorer non-food-producing countries, will lose. This theoretical result is confirmed by several empirical studies.
- **Food Aid** Even though the proposals from the Hong Kong talks were never ratified, and the Doha Round of negotiations is still ongoing, some recent progress has been made toward the goal of replacing food aid with efforts to increase production.

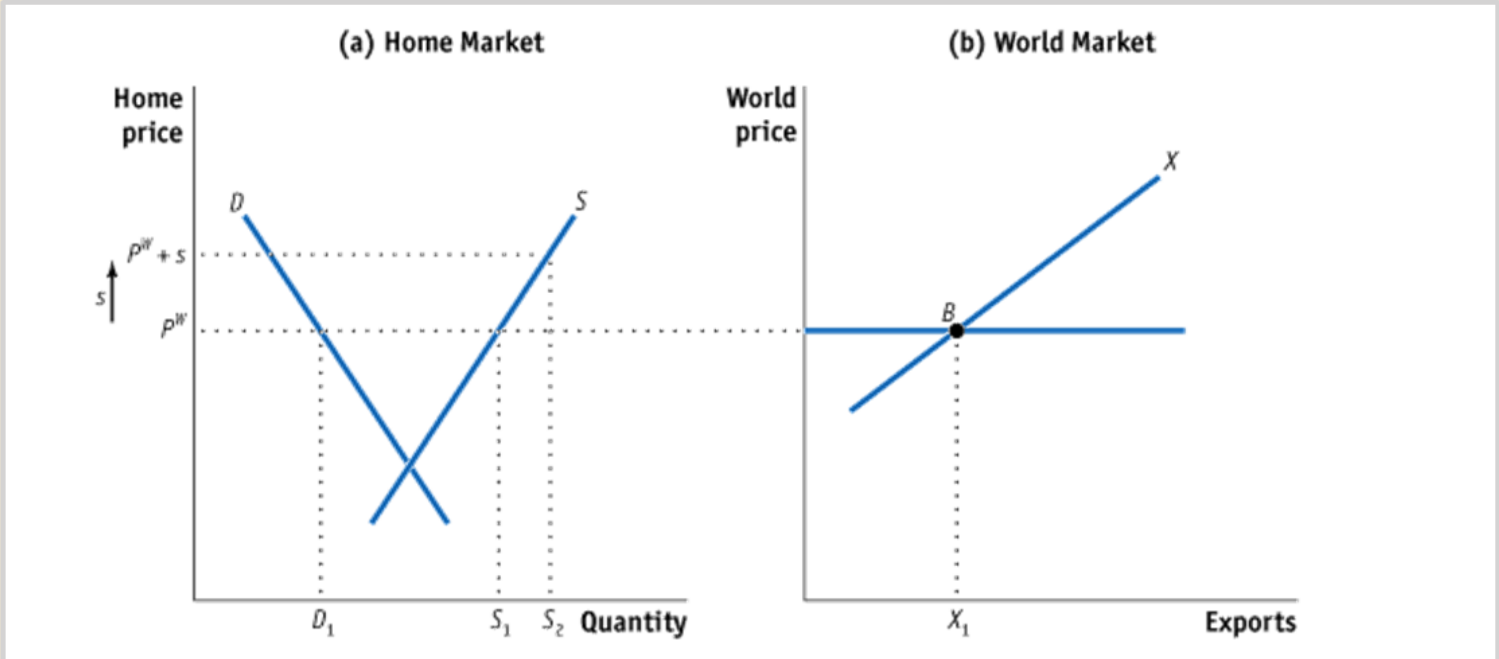
Agricultural Production Subsidies

- Suppose the government provides a subsidy of *s dollars for every unit* (e.g., *ton of sugar* in our example) that a Home firm produces. This is a **production subsidy** because it is a subsidy to every unit produced and not just to units that are exported.
- There are several ways that a government can implement such a subsidy.
  - The government might guarantee a minimum price to the farmer, for example, and make up the difference between the minimum price and any lower price for which the farmer sells.
  - Alternatively, it might provide subsidies to users of the crop to purchase it, thus increasing demand and raising market prices; this would act like a subsidy to every unit produced.

Agricultural Production Subsidies, cont'

Effect of a Production Subsidy in a Small Home Country

FIGURE 10-4 (1 of 2) **Production Subsidy for a Small Country**

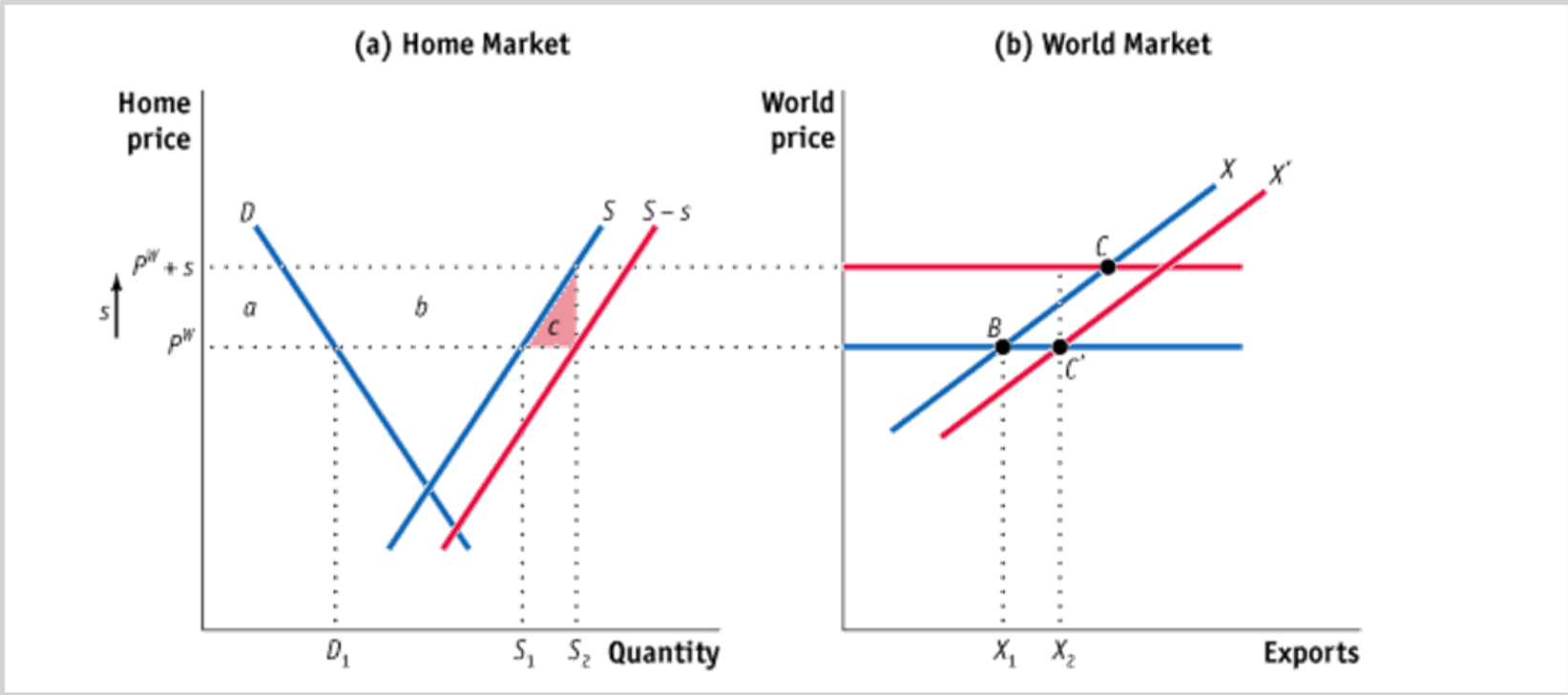


In panel (a), applying a production subsidy of *s* dollars per unit produced will increase the price that Home firms receive from P^W to $P^W + s$. This price rise leads to an increase in Home quantity supplied from S_1 to S_2. The consumer price at Home is not affected because the production subsidy does not distinguish between items sold at Home or exported (firms therefore continue to charge the world price at Home), so the quantity demanded stays at D_1.

Agricultural Production Subsidies, cont'

Effect of a Production Subsidy in a Small Home Country

FIGURE 10-4 (2 of 2) **Production Subsidy for a Small Country (continued)**

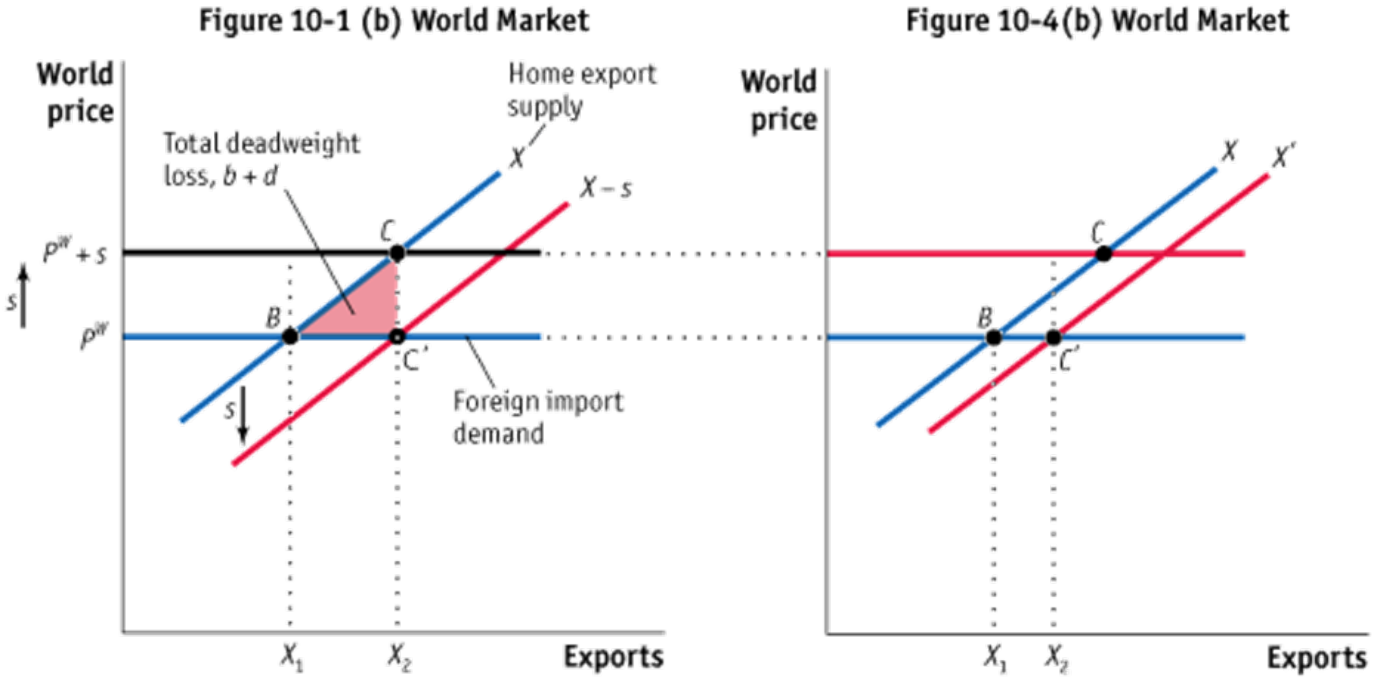


The deadweight loss of the subsidy for a small country is the area *c*. In panel (b), exports rise as a result of the production subsidy, from X_1 to X_2, though the increase in exports is less than for the export subsidy because, for the production subsidy, quantity demanded does not change at Home.

Agricultural Production Subsidies, cont'

- **Targeting Principle** Our finding that the deadweight loss is lower for the production subsidy makes it a better policy instrument than the export subsidy to achieve an increase in Home supply. This finding is an example of the targeting principle: *to achieve some objective, it is best to use the policy instrument that achieves the objective most directly*.
- There are many examples of using a targeting principle in economics:
  - Taxes on cigarettes and gasoline.
  - To use an example from this book, it is better to provide trade adjustment assistance directly to those affected, than to impost a tariff or quota.

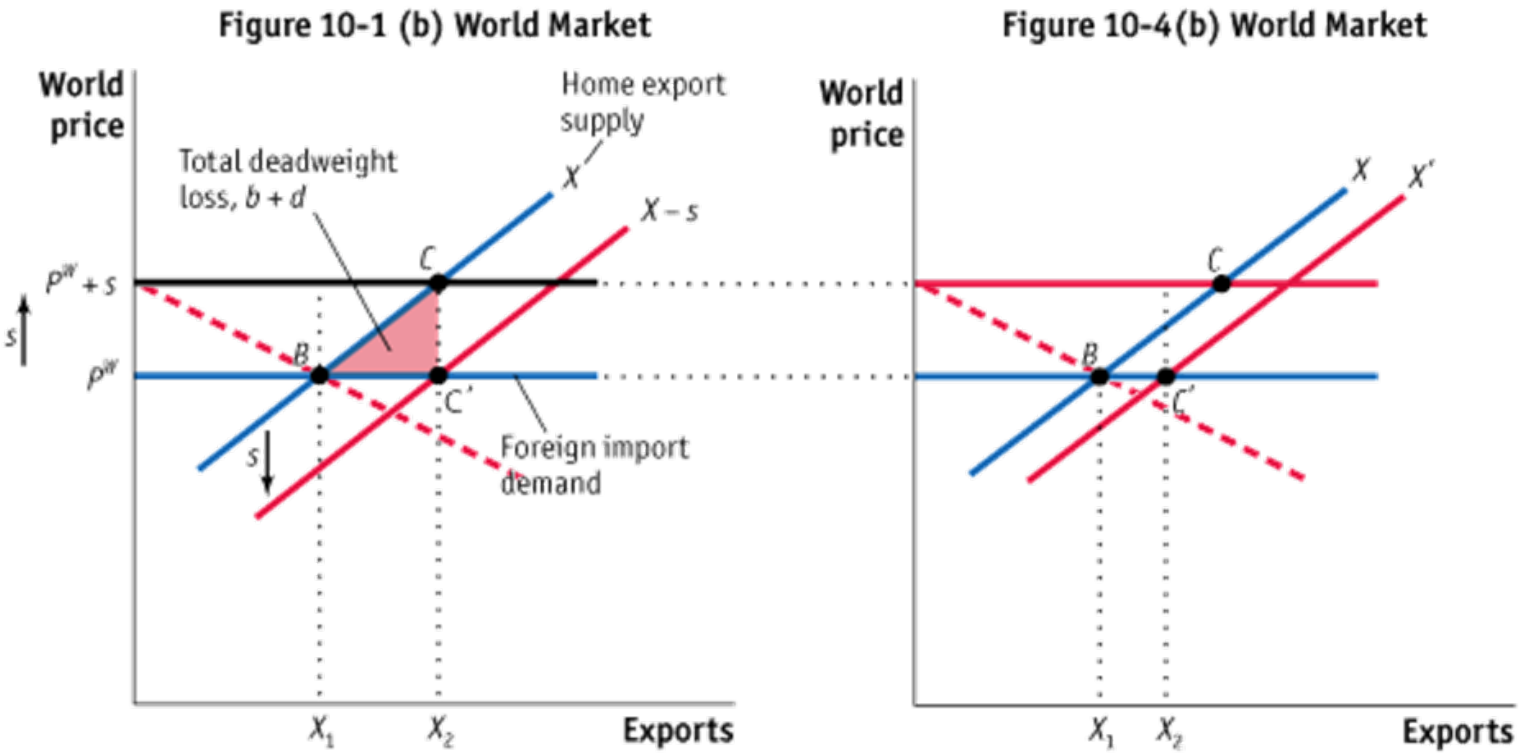
Agricultural Production Subsidies, cont'



With the export subsidy, the price for Home producers and consumers rose to $P^W + s$, so exports increased because of both the rise in quantity supplied and the drop in quantity demanded. As a result, the export subsidy shifted the Home export supply curve down by exactly the amount s in Figure 10-1. In contrast, with a production subsidy, exports rise only because Home quantity supplied increases so that export supply shifts down by an amount less than s in Figure 10-4.

Agricultural Production Subsidies, cont'

Effect of a Production Subsidy in a Large Home Country



If we drew a downward-sloping Foreign import demand curve in panel (b), then the increase in supply as a result of the production subsidy would lower the world price. But that drop in world price would be less than the drop that occurred with the export subsidy because the increase in exports under the production subsidy is less.

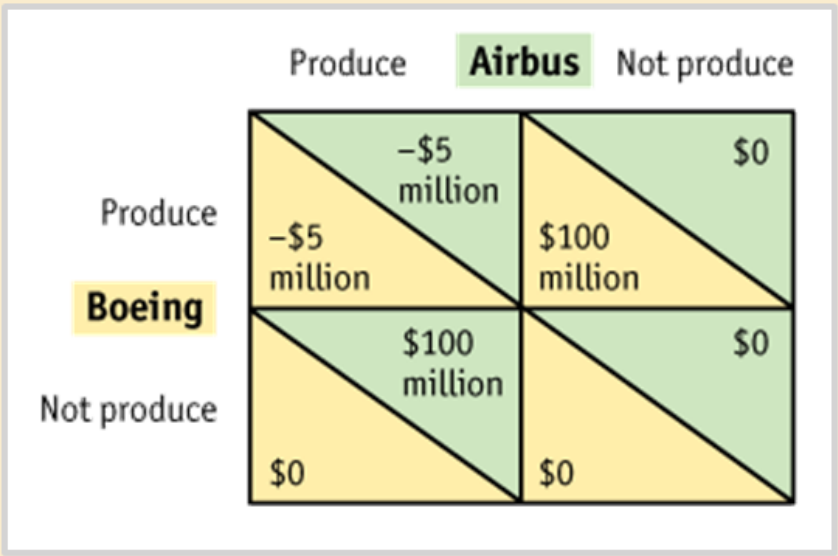
High-Technology Export Subsidies

- Governments subsidize high-technology industries because they may create benefits that spill over to other firms in the economy. That is, governments believe that high-tech industry produces a **positive externality**. This argument for a subsidy is similar to the infant industry argument used to justify protective tariffs.
- In addition to the spillover argument, governments and industries also argue that export subsidies might give a **strategic advantage** to export firms that are competing with a small number of rivals in international markets.
- To examine whether countries can use their subsidies strategically, we use the assumption of **imperfect competition**. Now we allow for two firms in the market, which is called a **duopoly**.
- To capture the strategic decision making of two firms, we use **game theory**, the modeling of strategic interactions (games) between firms as they choose actions that will maximize their returns.

High-Technology Export Subsidies, cont'

Payoff Matrix In Figure 10-5, we show a payoff matrix for Boeing and Airbus, each of which has to decide whether to produce the new aircraft.

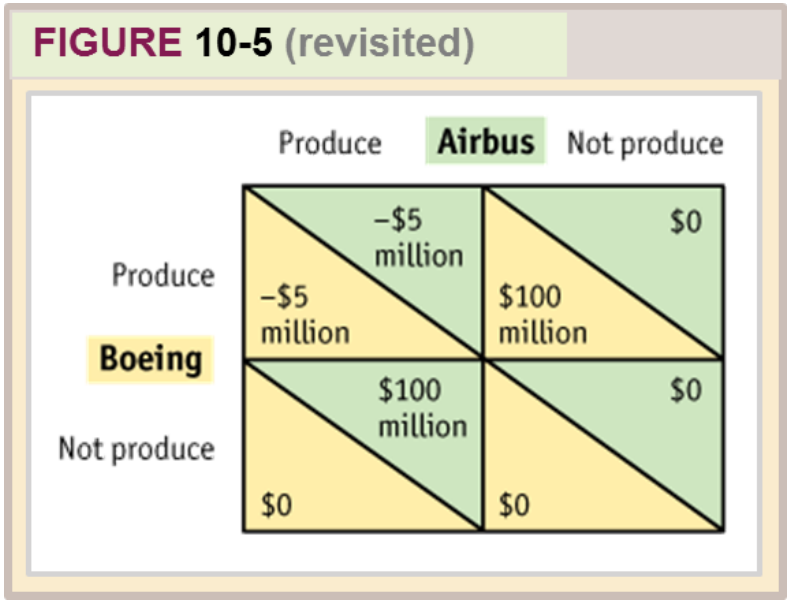
FIGURE 10-5 Payoff Matrix between Two Firms



The lower-left number in each quadrant shows the profits of Boeing, and the upper-right number shows the profits of Airbus. Each firm must decide whether to produce a new type of aircraft. A Nash equilibrium occurs when each firm is making its best decision, given the action of the other. For this pattern of payoffs, there are two Nash equilibria, in the upper-right and lower-left quadrants, where one firm produces and the other does not.

The idea of a **Nash equilibrium** is that each firm must make its own best decision, taking as given each possible action of the rival firm. When each firm is acting that way, the outcome of the game is a Nash equilibrium. The action of each player is the best possible response to the action of the other player.

High-Technology Export Subsidies, cont'



- **Best Strategy for Boeing** If Airbus produces, then Boeing is better off not producing. This finding proves that having both firms produce is not a Nash equilibrium. Boeing would never stay in production, since it prefers to drop out of the market whenever Airbus produces.
- **Best Strategy for Airbus** The decision illustrated in the lower-left quadrant, with Airbus producing and Boeing not producing, is a Nash equilibrium because each firm is making its best decision given what the other is doing.

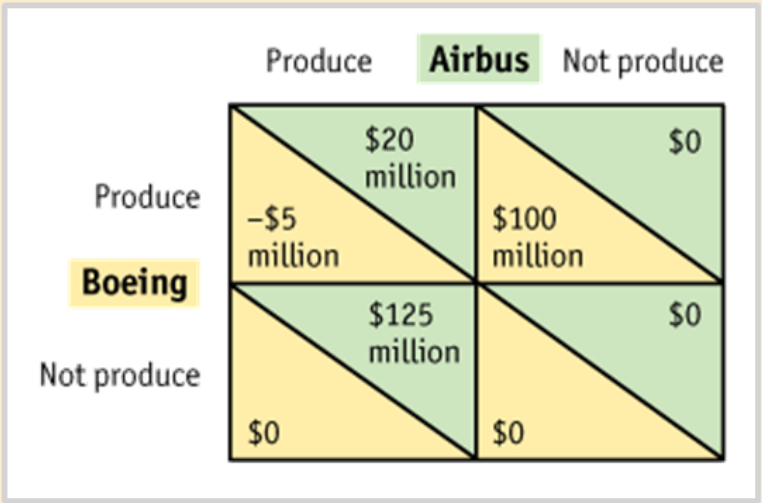
High-Technology Export Subsidies, cont'

- **Multiple Equilibria** The upper-right quadrant, with Boeing producing and Airbus not producing, is also a Nash equilibrium. When Boeing produces, then Airbus's best response is to not produce, and when Airbus does not produce, then Boeing's best response is to produce.
- When there are two Nash equilibria, there must be some force from outside the model that determines in which equilibrium we are. An example of one such force is the **first mover advantage**, which means that one firm is able to decide whether or not to produce before the other firm.
- Government interventions may also affect the equilibrium.

High-Technology Export Subsidies, cont'

Effect of a Subsidy to Airbus

FIGURE 10-6 Payoff Matrix with Foreign Subsidy



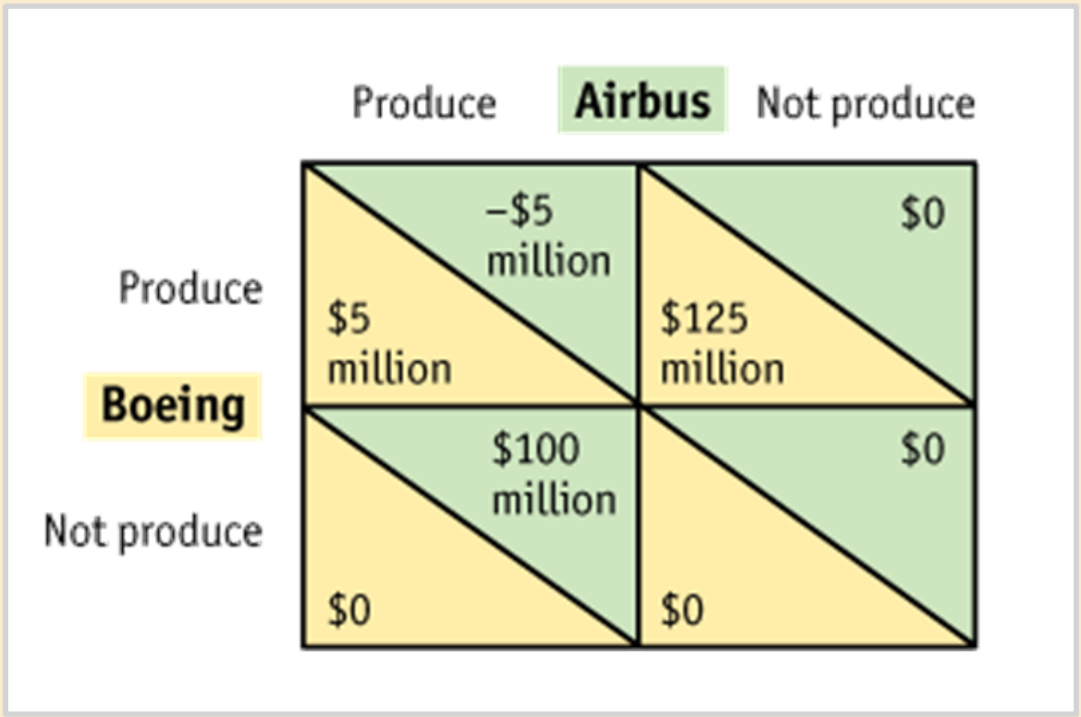
When the European governments provide a subsidy of $25 million to Airbus, its profits increase by that much when it produces a new aircraft. Now there is only one Nash equilibrium, in the lower-left quadrant, with Airbus producing but Boeing not producing. The profits for Airbus have increased from 0 to $125 million, while the subsidy cost only $25 million, so there is a net gain of $100 million in European welfare.

- **Best Strategy for Airbus** With the subsidy, Airbus now earns $20 million by producing instead of losing $5 million.
- **Best Strategy for Boeing** Boeing will want to drop out of the market. Once Boeing makes the decision not to produce, Airbus's decision doesn't change.

High-Technology Export Subsidies, cont'

Cost Advantage for Boeing

FIGURE 10-7 Another Payoff Matrix, with Boeing Cost Advantage

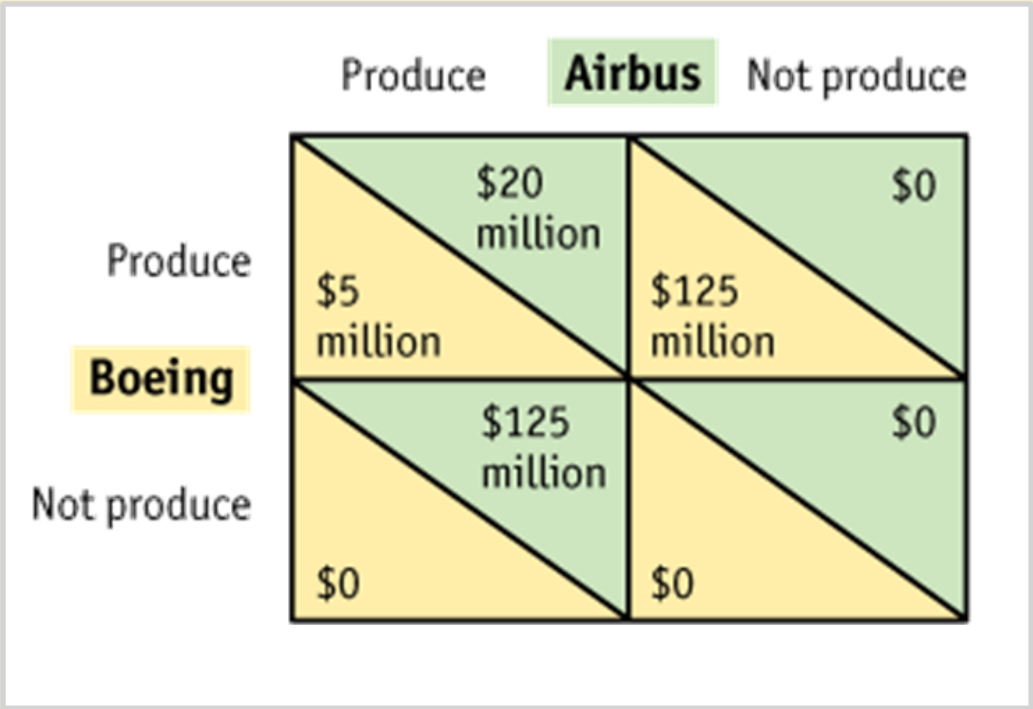


If Boeing has a cost advantage in the production of aircraft, the payoffs are as shown here. Boeing earns profits of $5 million when both firms are producing and profits of $125 million when Airbus does not produce. Now there is only one Nash equilibrium, in the upper-right quadrant, where Boeing produces and Airbus does not.

High-Technology Export Subsidies, cont'

Subsidy with Cost Advantage for Boeing

FIGURE 10-8 Another Payoff Matrix with Foreign Subsidy



When the European governments provide a subsidy of $25 million to Airbus, its profits increase by that much when it produces. Now the only Nash equilibrium is in the upper-left quadrant, where both firms produce. The profits for Airbus have increased from 0 to $20 million, but the subsidy costs $25 million, so there is a net loss of $5 million in European welfare.

High-Technology Export Subsidies, cont'

Subsidies for the large commercial aircraft industry include:

- Indirect subsidies that arise in the production of civilian and military aircraft;
- Direct subsidies for R&D;
- Subsidies of the interest rates that aircraft buyers pay when they borrow money to purchase aircraft;
- Others...

If both firms stay in the market and are subsidized by their governments, then it is unlikely that the subsidies are in the national interest of either the United States or the European Union; instead, the countries purchasing the aircraft gain because of the lower price, while the United States and Europe lose as a result of the costs of the subsidies.

Summary

Key Points:

1. An export subsidy leads to a fall in welfare for a small exporting country facing a fixed world price. The drop in welfare is a deadweight loss and is composed of a consumption and production loss, similar to an import tariff for a small country.

2. In the large-country case, an export subsidy lowers the price of that product in the rest of the world. The decrease in the export price is a terms-of-trade loss for the exporting country. Therefore, the welfare of the exporters decreases because of both the deadweight loss of the subsidy and the terms-of-trade loss. This is in contrast to the effects of an import tariff in the large-country case, which generates a terms-of-trade gain for the importing country.

Summary

Key Points:

3. Export subsidies applied by a large country create a benefit for importing countries in the rest of the world, by lowering their import prices. Therefore, the removal of these subsidy programs has an adverse affect on those countries. In fact, many of the poorest countries are net food importers that will face higher prices as agricultural subsidies in the European Union and the United States are removed.

4. Production subsidies to domestic producers also have the effect of increasing domestic production. However, consumers are unaffected by these subsidies. As a result, the dead-weight loss of a production subsidy is less than that for an equal export subsidy, and the terms-of-trade loss is also smaller.

Summary

5. It is common for countries to provide subsidies to their high-technology industries because governments believe that these subsidies can create a strategic advantage for their firms on international markets. Because these industries often have only a few global competitors, we use game theory (the study of strategic interactions) to determine how firms make their decisions under imperfect competition.
6. A Nash equilibrium in a game is a situation in which each player is making the best response to the action of the other player. In a game with multiple Nash equilibria, the outcome can depend on an external factor, such as the ability of one player to make the first move.
7. Export subsidies can affect the Nash equilibrium of a game by altering the profits of the firms. If a subsidy increases the profits to a firm by more than the subsidy cost, then it is worthwhile for a government to undertake the subsidy. As we have seen, though, subsidies are not always worthwhile unless they can induce the competing firm to exit the market altogether, which may not occur.